Juhyeon Park

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Education

2023 - present **Seoul National University**
Ph.D Candidate in Interdisciplinary Program in Artificial Intelligence
(Advisor: Prof. Taesup Moon)

2017 - 2023 **Seoul National University**
B.S. in Electrical and Computer Engineering (Minor: Statistics) (GPA 3.94/4.3)
Thesis: CIF: Cleansing data with influence score for fairness

Research Interest

- Brain + AI (Alzheimer's Disease Diagnosis, Brain Decoding, Multimodality)
  "Can AI help understand the brain? Can understanding of the brain advance AI?"
- AI Safety (Risks from model malfunctioning: debiasing, fairness, and privacy leakage)
- AI Evaluation *"How can we faithfully evaluate the model's ability?"*

Publications

(*: Equal Contribution, J: Journal, C: Conference, W: Workshop)

[**C4, W2**] Sumin Yu*, **Juhyeon Park*** and Taesup Moon (2026) "PopResume: Causal Fairness Evaluation of LLM/VLM Resume Screeners with Population-Representative Dataset". In: *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP) Oral presentation, NeurIPS 2026 TAE Workshop.*

Contributions: Proved the main lemma, implemented the path-specific effect estimation framework, collected seed data for data generation, and performed data imputation.

[**C3**] Peter Yongho Kim*, **Juhyeon Park***, Jungwoo Park, Jubin Choi, Jungwoo Seo, Jiook Cha, and Taesup Moon (2026) "Can Natural Image Autoencoders Compactly Tokenize fMRI Volumes for Long-Range Dynamics Modeling?" In: *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

Contributions: Preprocessed fMRI data, co-developed a novel tokenization scheme, identified the best-performing architecture, and designed a new task to study the link between long-range fMRI dynamics and task performance.

[**J1**] **Juhyeon Park***, Donggyu Lee* and Taesup Moon (2026) "Uncovering Group Robustness Issue in Early Diagnosis of MCI to AD Conversion and A Simple Solution: Decoupled Classifier with Adaptive Linear Modulation". In: *Biomedical Signal Processing and Control.*

Contributions: Preprocessed the data, co-developed the main algorithm, led prognosis-task experiments, and conducted interpretation-related analyses.

[**C2**] **Juhyeon Park***, Peter Yongho Kim*, Jiook Cha, Shinjae Yoo, and Taesup Moon (2026) "SEED: Towards More Accurate Semantic Evaluation for Visual Brain Decoding". In: *International Conference on Learning Representations (ICLR).*

Contributions: Developed Cap-Sim, designed a human evaluation framework for meta-evaluation, and identified failure modes of current visual decoding models.

[**W1**] **Juhyeon Park***, Peter Yongho Kim*, Jungwoo Park*, Jubin Choi, Jungwoo Seo, Jiook Cha, and

Taesup Moon (2025) "Processing fMRI Brain Signals Using Latents from Natural Image Autoencoders". In: *Foundation Models for the Brain and Body NeurIPS 2025 Workshop* Oral presentation.

Contributions: Preprocessed fMRI data, co-developed a novel tokenization scheme and explored architectures to identify the best-performing design.

[**C1**] **Juhyeon Park***, Seokhyeon Jeong* and Taesup Moon (2025) "TLDR: Text Based Last-layer Retraining for Debiasing Image Classifiers". In: *The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.

Contributions: Developed the main algorithm, proved the key lemma, and led all experiments.

Preprints

(*: Equal Contribution, P: Preprint)

[**P3**] **Juhyeon Park***, Sumin Yu*, and Taesup Moon (2026) "ConCI: A Controlled Counterfactual Benchmark for Contextual Integrity in LLMs". In: *Preprint*.

[**P2**] **Juhyeon Park**, Yeonwoo Kim, Peter Yongho Kim, Yansen Wang, Mingqing Xiao, Dongqi Han, Dongsheng Li, and Taesup Moon (2026) "Natural Image Autoencoder-Based fMRI Representations for Trait and State Prediction". In: *Preprint*.

[**P1**] Jin Park*, Jung Yoon Hwang*, Sangwon Jung, **Juhyeon Park**, and Taesup Moon (2026) "SparSlice: Sparse Latent Slice Discovery for Vision Models". In: *Preprint*.

Projects

Development and Demonstration of an AI Care Service for Dementia Prevention Based on a Large-Scale Brain–Omics–Language Foundation Model — Jul 2025 – Present

Decoding Emotion-Contextualized Perception during Brain–External Environment Interaction Based on an Animal–Human Multiscale Neuroscience Foundation Model — Aug 2023 – Present

Development of Explainable AI and Robust Data Processing Technologies Incorporating Cross-Modal Correlations in Neuroscience Data, NRF of Korea — Apr 2022 – Dec 2023

Experience

Research Intern, Microsoft Research Asia - AI/ML Group — Mar 2026 – Sep 2026

(Manager: Dongsheng Li, Mentor : Yansen Wang, Mingqing Xiao, Dongqi Han)

Intern, Naver CLOVA - Image Vision Team — Jul 2022 – Dec 2022

Developed a lightweight algorithm for human matting and a platform for automated model evaluation and leaderboarding. (Manager: Youngjoon Yoo, Mentor: Meongyeon Lee)

Research Intern, M.IN.D Lab — Jul 2021 – Feb 2023

Developed an algorithm to enhance classifier fairness by cleansing training data using influence scores.

Honors & Awards

Outstanding Paper Award KCC 2026 XAI Workshop — June 2026

For "PopResume: Causal Fairness Evaluation of LLM/VLM Resume Screeners

with Population-Representative Dataset"

Hyundai Motor Chung Mong-Koo Foundation Scholarship Sep 2023 - Present
Hyundai Motor Chung Mong-Koo Foundation

SNU Merit-based Scholarship Spring 2023
Dept. of Electrical and Computer Engineering, Seoul National University

SNU Merit-based Scholarship Spring 2021
Dept. of Electrical and Computer Engineering, Seoul National University

HYUNSONG Educational & Cultural Foundation Scholarship Mar 2018 – Feb 2023
HYUNSONG Educational & Cultural Foundation

Acadmeic Services

Reviewer

AAAI (26', 27'), NeurIPS (26' Top Reviewer), ICLR (27'), ACL ARR (26' Oct)